

Last name _____

First name _____

LARSON—MATH 310—CLASSROOM WORKSHEET 16
Matrix-Matrix Multiplication.

Review

1. (**Systems of Linear Equations**, Chp. 8) Rewrite the following linear system in matrix notation.

$$\begin{array}{rcrcrcrcrcrl} x & + & 2y & + & 3z & = & 6 \\ 2x & + & 3y & + & 4z & = & 9 \\ 3x & + & 4y & + & 5z & = & 10 \end{array}$$

2. Given matrices A , B , what is the definition of AB ?
3. Find AB using the *definition* of matrix multiplication.

Matrix Multiplication (Chp. 10)

Let

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 3 & 1 \\ 3 & 1 & 2 \end{bmatrix}$$

$$B = \begin{bmatrix} 1 & 2 \\ 2 & 3 \\ 3 & 1 \end{bmatrix}$$

1. Find AB .

2. Find $(AB)^T$.

3. Find A^T .

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 3 & 1 \\ 3 & 1 & 2 \end{bmatrix}$$

$$B = \begin{bmatrix} 1 & 2 \\ 2 & 3 \\ 3 & 1 \end{bmatrix}$$

4. Find B^T .

5. Find $B^T A^T$. Check that $(AB)^T = B^T A^T$.

6. Find AB using the *column interpretation* of matrix multiplication.

7. Matrices A and B define linear functions. What linear function does AB define?

8. Find A^2 .